Pre-Lab 5 | Forced vibration

Submit one Word or PDF file on Canvas before your own Lab 5 section. Include the requested figures and brief answers below. Use the supplied Colab notebook or MIE402_PreLab5.m; no additional programming is required. Export figures directly and define variables in equations you use.

A. Predict, then run

For an illustrative system use m = 1 kg, c = 1 N·s/m and k = 100 N/m. Calculate natural angular frequency $\omega_n = \sqrt{(k/m)}$, damping ratio $\zeta = c/(2\sqrt{(mk)})$, and damped angular frequency $\omega_d = \omega_n\sqrt{(1-\zeta^2)}$. Convert angular frequencies to Hz. Explain the difference between these frequencies and the imposed drive frequency.

B. Short impact

Run the unit-impulse response for impulse J = 1 N·s, starting from rest. Include the displacement plot. What happens to amplitude and frequency if J doubles? Explain using the impulse response equation. The laboratory tap is not a measured J unless force versus time is recorded.

C. Harmonic force and startup

Use F(t) = 1 cos(4πt) N and initial displacement 0.1 m with initial velocity zero. Include the full-response/steady-state plot. Calculate the drive frequency in Hz. Explain why the two curves differ early and approach each other later. At late time, does the cart oscillate at f_n or the drive frequency?

D. Connect the simulation to the actual apparatus

The laboratory motor moves one spring end. Use k_1 = 45 N/m, k_2 = 55 N/m, m = 1 kg, c = 1 N·s/m and endpoint amplitude Y = 0.01 m. Include the predicted X/Y frequency-response curve. State its low-frequency limit and explain why it is not 1. The notebook compares transfer-response and state-equation results for this input.

E. Prepare your bench log

State how you will measure each of: total moving mass, spring stiffnesses, input peak amplitude and drive frequency. Convert a 0.024 m peak-to-peak input to Y, and 72 rpm to Hz and rad/s. Explain why changing magnet count requires a matching damping estimate. Give one way to recognize that a harmonic record is not yet steady.

Bring: your spring IDs and Lab 4 calibration results if you will reuse those springs. Save your pre-lab figures and written explanations in one document. The supplied numerical values are not substitutes for your group’s measurements.